Chapter 1: Coupled Simulation

Now we couple locally mass conserved MFEM and FVM method together in this chapter and perform coupled numerical simulation for partially molten mantle. We discuss implementation details and coupling strategy combining MFEM, FVM and eutectic phase package. The remaining uncertainty of porosity ϕ will be fulfilled by phase package evaluation. Before we throw ourselves into the fully coupled problem. We test another porosity evolution scenerio with a more straightforward prescribed melting. Then we will show an example combining MFEM, FVM and eutectic phase package alltogether and perform coupled computation. We start with pseudo 1D scenerios and then proceed to perform full 2D simulation.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ be the cell-averaged solution to the transport system (**??**) at time t^n. Let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ denote the solution to the Darcy-Stokes system at the same time step t^n. Moreover, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n given the transport result U^n.

For each time step t^n, we evaluate ML-WENO reconstructed dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$. The eutectic phase package gives corresponding phase parameters $\mathcal{W}^n$ according to the pointwise values. Once we know $\mathcal{W}^n$, we solve a fixed in time Darcy-Stokes system at this time step t^n for $\mathcal{V}^n$. The cell-averaged solution $\mathcal{U}^{n+1}$ of the next time step t^{n+1} is then obtained by forward marching the system explicitly. We formally formulate an overall explicit coupled solver as shown in algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

1: Initialize $\mathcal{U}^0$, comptute $\mathcal{W}^0$ with eutectic phase package.

2: Compute ϕ^0 in $\mathcal{W}^0$, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^0$.

3: **for** Time level $n = 1, 2, \ldots, n_{\max} - 1$ **do**

4: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

5: Compute ϕ^n in W^n, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^{n+1}$.

6: Given $\mathcal{W}^n$ and $\mathcal{V}^{n+1}$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

7: Explicitly marching forward the transport system (**??**) to next time $n+1$ for $\mathcal{U}^{n+1}$.

8: **end for**

1.1.1 Discontinuous Data

In the coupled computation, we obtain different data type that require different and careful investigation.

1.2 Pseudo One-Dimensional Column Tests

We first test our coupled algorithm with a pseudo-1D column problem.

1.2.1 Boundary condition

1.2.2 Prescribed Melting Case

Before fully coupled computation, we illustrate a simpler model coupling case coupling MFEM and FVM solver. Instead of deciding porosity with phase package computation, we prescribe the melting rate or the change of porosity as a fixed function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.1}$$

We write conservation of liquid mass with the prescribed melting term Γ (1.1) as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.2}$$

where diffusion effect is being dropped for simplicity.

We perform a quasi-1D column test on a 2×40 rectangular physical domain $\Omega \in [-0.05, 0.05] \times [-0.4, 0]$. We prescribe the following boundary conditions for MFEM and FVM solver.

- Bottom side;
  - MFEM. We prescribe solid velocity at the bottom of the physical domain, indicating supplying of mantle.

    $$\check{\mathbf{v}}_s = \mathbf{v}_{\text{upwelling}}.$$

    We prescribe zero liquid velocity indicating no melting penetrating bottom of the computational domain;
  - FVM. We prescribe a fixed inflow boundary condition, supplying composition and enthalpy into system.
- Left and right side;
  - MFEM. We prescribe natural boundary conditon giving zero traction to the Stokes equation in tangential direction. We prescribe zero flux for dofs in normal direction for the Stokes equation and Darcy equation.
  - FVM. We prescribe noflow (zero flux) condition.
- Top side;

- MFEM. We prescribe natural boundary conditoin given zero traction and zero pressure to Stokes and Darcy part respectively.
- FVM. We set free outflow boundary if it is recognized as outflow. We set zero if the velocity reversed to inflow boundary condition.

For simplicity and efficiency in computation, we use 1×3 and 1×2 for our ML-WENO(3,2) stencils, since we don't need resolution and high order accuracy in x direction. We show two different melting source scenarios. One of which is a piece wise constant stair function:

$$\Gamma_1(z) = \begin{cases} 1e-3, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.3}$$

and one with melting source trapped in the middle of the melting column as

$$\Gamma_2(z) = \begin{cases} 0, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.4}$$

We assign different initial conditions to these two scenarios as

$$\phi_1(z,0) = 0.0, \quad z \in [-0.4, 0.0], \tag{1.5}$$

and

$$\phi_2(z,0) = \begin{cases} 1e-2, & -0.2 < z \leq 0, \\ 0, & -0.4 \leq z \leq -0.2. \end{cases} \tag{1.6}$$

1.2.3 Phase Transition Case

1.3 Two-Dimensional Test